Worksheet 6-8 & 6-7 B Basic Integrals and the FTC #1
AP Calculus Name_____

Directions: Evaluate the following.

1. $\int_0^5 (4x-3)\,dx =$

2. $\int (x^{-3}+5x^3)\,dx =$

3. $\int_{\frac{\pi}{3}}^{\pi} \cot\theta \sin\theta \;\, d\theta \;=$

4. $\int_1^5 \frac{xe^x+4x+2}{x}\,dx$

5. $\int_1^9 \sqrt{x}\,dx \;=$

6. $\int \left(e^x+\frac{4}{x}+2x^3\right)dx =$

7. $\int_{-1}^4 (x^2-2x+1)\,dx =$

8. $\int \left(\frac{2x-6}{x^2-3x}\right)dx =$

9. $\int_0^{\frac{\pi}{2}} (\cos x+\sin x)\,dx =$

10. $\int_1^3 \frac{-4}{x^2}\,dx =$

11. $\int (1-\sec^2 t)\,dt =$

12. $\int \left(\frac{1-x}{x}\right)dx =$

13. $\int (2\sec\theta\tan\theta)\ d\theta =$

14. $\int_{\frac{\pi}{3}}^{\frac{5\pi}{6}} -2\sin x\ \mathrm{dx} =$

15. $\int \left(\frac{x^{\frac{5}{2}} + \sqrt{x} - 1}{\sqrt{x}}\right) dx =$

16. $\int \left(y^2\sqrt{y}\right) dy =$

17. $\int_1^9 \frac{1}{\sqrt{x}} dx =$

18. $\int_{-1}^{1} (3x)^2\, dx =$

19. $\int_1^{e^2} \frac{1}{x} dx =$

20. $\int_{-2}^{2} x^3\, dx =$

21. $\int_1^3 \frac{2x^2 - 7x - 15}{x - 5} dx =$

22. $\int \left(\frac{5x^2e^{-x} - 2}{e^{-x}}\right) dx =$

23. $\int_{-1}^{3} (x^2 + 3)\, dx =$

24. $\int -3\, dx =$

25. $\int_2^5 ((x-2)(3x+1))\, dx =$

26. $\int_{-2}^{4} \left(\frac{x^2 - 25}{x + 5}\right) dx =$